

On Mutual Information Maximization in Linear Sensing with Arbitrary Channel Priors

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Abstract—This paper considers multiple-input single-output (MISO) channel estimation under a linear sensing framework from a mutual information (MI) maximization perspective. Unlike conventional channel estimation approaches that optimize mean-square error, we treat the channel vector as an information-bearing source that implicitly encodes environmental parameters. In particular, we design a downlink sensing matrix that directly maximizes the MI between the received measurements and the underlying channel vector. We study general linear sensing matrices that may be under-determined and enable cross-channel power coupling. We derive a necessary optimality condition for sensing matrix design and establish its global optimality under Gaussian inputs. Based on this condition, we develop an iterative algorithm for optimizing the sensing matrix under arbitrary input distributions. Numerical results demonstrate that significant enhanced MI over conventional power allocation-based designs.

I. INTRODUCTION

Reliable communication between a base station (BS) and user equipment (UE) depends on effective acquisition of Channel State Information (CSI). Traditional channel estimation methods focus on minimizing the fidelity-centric mean-squared error (MSE) [1], which is well suited for data decoding but not always aligned with the objectives of emerging applications. In many cases, the primary concern is not accurate channel reconstruction, but the amount of task-relevant information conveyed by the channel observations. This is particularly evident in Integrated Sensing and Communication (ISAC) systems, where channel responses implicitly encode environmental information [2]. A similar paradigm arises in unmanned aerial vehicle (UAV)-enabled networks, where UAVs act as mobile relays or sensing platforms under strict power and latency constraints. In such settings, exhaustive channel sounding is impractical, and efficient sensing matrix design becomes essential to extract task-oriented information, such as location or motion of the objects, from a limited number of measurements [3].

This work is supported in part by the General Research Fund (project number 14202421, 14214122, 14202723, 14207624), Area of Excellence Scheme grant (project number AoE/E-601/22-R), and NSFC/RGC Collaborative Research Scheme (project number CRS_HKUST603/22, CRS_HKU702/24), all from the Research Grants Council of Hong Kong; in part by the National Natural Science Foundation of China (Project number 62571336), in part by the Guangdong Basic and Applied Basic Research Foundation (Project number 2025A1515010249), and in part by the Shenzhen University 2035 Program for Excellent Research (Grant No. 2024C010).

Motivated by these considerations, some have proposed adopting an information-theoretic viewpoint and design sensing strategies that maximize the mutual information (MI) between the channel vector and its noisy compressed measurements. MI directly quantifies how much uncertainty about the channel is resolved by the observations, and therefore provides a principled measure of the effectiveness of channel sensing [4], [5]. Interestingly, the scheme in which the sensing matrix aligns with the channel eigenmodes and allocates power according to the water-filling (WF) principle [6] is exactly the optimal solution for mutual information maximization in the classical linear Gaussian model under an average power constraint. This result serves as a fundamental benchmark for sensing and precoding design.

Although the WF solution is optimal under the Gaussian input assumption, it is rarely employed in practical systems due to their incompatibility with many realistic channel or signal models. In practice, channel vectors often exhibit structured, non-Gaussian distributions, arising from propagation environments, antenna geometries, or empirical measurements. For instance, DeepMIMO [7] provide a large collections of high-dimensional channel realizations, that cannot be adequately described by simple parametric Gaussian models. Without the Gaussian input assumption, the classical WF solution is no longer optimal, motivating the design of general information-maximizing sensing matrices [6]. To address information-maximizing precoder design under arbitrary input distributions, [4] exploits the gradient characterization of mutual information [8] to derive a necessary optimality condition for such precoders. This line of work establishes a general framework for extending mutual-information-based transceiver optimization beyond the Gaussian signaling assumption.

In contrast to [4], which focuses on precoder optimization for data transmission over a fixed channel realization, this paper addresses a different yet related problem: **the design of power-constrained sensing matrices for channel estimation**. Specifically, we consider the optimization of the sensing matrix by directly maximizing the mutual information between the random channel vector and its noisy linear measurements. This shift in perspective leads to an optimization problem over potentially under-determined sensing matrices.

Building on the I-MMSE gradient identity [4], we formulate MI-driven sensing matrix design for arbitrary channel priors.

We first derive a necessary stationarity condition for the power-constrained sensing matrix. Under Gaussian channel priors, this condition reduces to the classical Principle Component Analysis (PCA) plus WF structure, recovering the known optimal solution. For general empirical or discrete channel priors, we propose a normalized I-MMSE gradient-ascent algorithm in which the MMSE matrix is estimated by Monte Carlo posterior averaging [7]. Numerical results on DeepMIMO channel samples show that the proposed design improves empirical MI over covariance-eigenmode power allocation baselines [6].

II. SYSTEM MODEL AND PRELIMINARIES

We consider a linear downlink sensing model given by

$$\mathbf{y} = \mathbf{A}\mathbf{h} + \mathbf{w}, \quad (1)$$

where $\mathbf{h} \in \mathbb{C}^n$ denotes the channel vector, and n is the number of receiving antennas, $\mathbf{A} \in \mathbb{R}^{m \times n}$ is the sensing matrix with $m \leq n$, and $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_M)$ is additive white Gaussian noise. The sensing matrix is subject to a total power (denoted by P) constraint

$$\|\mathbf{A}\|_F^2 = \text{Tr}(\mathbf{A}\mathbf{A}^H) \leq P. \quad (2)$$

The channel vector $\mathbf{h}$ is modeled as a random vector with prior distribution $p(\mathbf{h})$, which may be arbitrary. Our objective is to design $\mathbf{A}$ so as to maximize the MI

$$\max_{\text{Tr}(\mathbf{A}\mathbf{A}^H) \leq P} I(\mathbf{h}; \mathbf{y}). \quad (3)$$

If $\mathbf{h}$ is zero-mean Gaussian with covariance matrix $\Sigma_{\mathbf{h}}$, the MI admits the closed-form expression

$$I(\mathbf{h}; \mathbf{y}) = \log_2 \det \left(\mathbf{I}_M + \frac{1}{\sigma^2} \mathbf{A} \Sigma_{\mathbf{h}} \mathbf{A}^T \right). \quad (4)$$

In this case, the optimal sensing matrix aligns its row space with the dominant eigenmodes of $\Sigma_{\mathbf{h}}$, with power allocated according to the classical WF policy.

For arbitrary channel distributions, the expression for $I(\mathbf{h}; \mathbf{y})$ will be more complicated, and the optimal sensing strategy generally deviates from WF, therefore, we admit the fixed-point iteration based sensing matrix optimization method proposed by [4], which will be detailed in Section III.C.

III. MI-DRIVEN SENSING MATRIX DESIGN

This section develops a sensing matrix design framework based on MI maximization. We first derive a matrix-form necessary optimality condition by characterizing the gradient of MI with respect to the sensing matrix, yielding an I-MMSE-type relation applicable to general input distributions. We then specialize this condition to the Gaussian input case, where it reduces to a closed-form structure equivalent to principal component analysis combined with classical WF, as shown in Fig. 1.

A. Necessary Optimality Condition via the I-MMSE Gradient

Consider the Lagrangian of (3)

$$\mathcal{L}(\mathbf{A}, \mu) = I(\mathbf{h}; \mathbf{y}) - \mu (\|\mathbf{A}\|_F^2 - P), \quad (5)$$

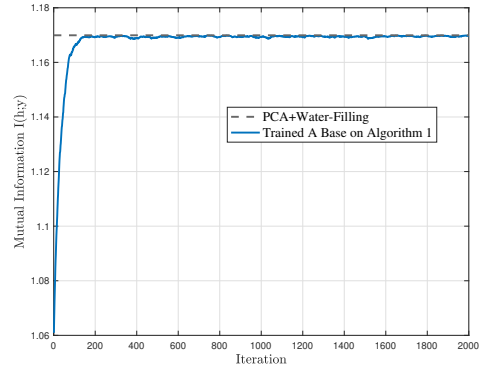


Fig. 1. Generalization of fixed-point iteration to Water-Filling+PCA solution with Gaussian Input.

where $\mu \geq 0$.

Lemma 1 (Necessary I-MMSE Optimality Condition). *Let $\mathbf{A}^*$ be a stationary point of (3). Then $\mathbf{A}^*$ necessarily satisfies*

$$\mathbf{A}^* \mathbf{E} = \mu \sigma^2 \mathbf{A}^*, \quad (6)$$

where

$$\mathbf{E} \triangleq \mathbb{E} [(\mathbf{h} - \mathbb{E}[\mathbf{h} | \mathbf{y}])(\mathbf{h} - \mathbb{E}[\mathbf{h} | \mathbf{y}])^T] \quad (7)$$

is the MMSE matrix and $\mu \geq 0$ is the Lagrange multiplier. This condition is necessary but not sufficient for global optimality. Moreover, if $\mathbf{A}^* \neq \mathbf{0}$, then $\mu > 0$ and $\|\mathbf{A}^*\|_F^2 = P$.

Proof. From the I-MMSE gradient formula [8],

$$\nabla_{\mathbf{A}} I(\mathbf{h}; \mathbf{y}) = \frac{1}{\sigma^2} \mathbf{A} \mathbf{E}. \quad (8)$$

The KKT stationarity condition $\nabla_{\mathbf{A}} \mathcal{L}(\mathbf{A}, \mu) = \mathbf{0}$ yields

$$\frac{1}{\sigma^2} \mathbf{A}^* \mathbf{E} - \mu \mathbf{A}^* = \mathbf{0}, \quad (9)$$

which implies (6). If $\mathbf{A}^* \neq \mathbf{0}$, complementary slackness enforces $\mu > 0$ and hence $\|\mathbf{A}^*\|_F^2 = P$. $\square$

Fixed-point form: Equation (6) implies colinearity between $\mathbf{A}^*$ and $\mathbf{A}^* \mathbf{E}$, leading to the normalized fixed-point update

$$\mathbf{A}^* = \sqrt{P} \frac{\mathbf{A}^* \mathbf{E}}{\|\mathbf{A}^* \mathbf{E}\|_F}. \quad (10)$$

B. Gaussian Prior Specialization: Generalization to PCA and WF

For Gaussian inputs, the global maximizer of (3) reduces to the PCA+WF structure shown in Corollary 1. This does not imply that the normalized iteration (12) is globally convergent. Rather, (12) should be viewed as a normalized I-MMSE gradient-ascent iteration whose limit points, when they exist, satisfy a stationary version of the necessary condition.

Corollary 1 (Gaussian Prior: PCA and WF as the Global Solution). *Assume the channel vector $\mathbf{h} \sim \mathcal{N}(\mathbf{0}, \Sigma_{\mathbf{h}})$ and consider the linear Gaussian model (1) under the Frobenius power constraint (2). Let*

$$\Sigma_{\mathbf{h}} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^H, \quad \mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n),$$

where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$. Then a globally optimal sensing matrix can be written as

$$\mathbf{A}^* = \mathbf{U} [\text{diag}(d_1^*, \dots, d_m^*), \mathbf{0}] \mathbf{V}^H,$$

where $\mathbf{U}$ is an arbitrary unitary matrix. The power allocated to the eigenmodes of $\Sigma_{\mathbf{h}}$ follows the WF rule

$$(d_i^2)^* = \left(\nu - \frac{\sigma^2}{\lambda_i} \right)_+, \quad \sum_{i=1}^m (d_i^2)^* = P, \quad (11)$$

where $\nu > 0$ is chosen to satisfy the power constraint, and modes with $\lambda_i = 0$ receive zero power. Moreover, this globally optimal solution satisfies the necessary condition (6).

Proof. Let

$$\mathbf{Q} = \mathbf{A}^H \mathbf{A}.$$

Then $\mathbf{Q} \succeq \mathbf{0}$, $\text{rank}(\mathbf{Q}) \leq m$, and $\text{Tr}(\mathbf{Q}) = \|\mathbf{A}\|_F^2 \leq P$. By Sylvester's determinant identity, maximizing (4) is equivalent to optimizing over $\mathbf{Q}$ with these constraints.

Let $q_1 \geq q_2 \geq \dots \geq q_n \geq 0$ denote the eigenvalues of $\mathbf{Q}$, where $q_i = 0$ for $i > m$. For fixed eigenvalues $\{q_i\}$, the determinant inequality for products of positive semidefinite matrices gives

$$\det\left(\mathbf{I}_n + \frac{1}{\sigma^2} \Sigma_{\mathbf{h}}^{1/2} \mathbf{Q} \Sigma_{\mathbf{h}}^{1/2}\right) \leq \prod_{i=1}^n \left(1 + \frac{\lambda_i q_i}{\sigma^2}\right),$$

with equality when $\mathbf{Q}$ and $\Sigma_{\mathbf{h}}$ are simultaneously diagonalizable and their eigenvalues are paired in the same decreasing order. Hence, the optimal right singular vectors of $\mathbf{A}$ align with the eigenvectors of $\Sigma_{\mathbf{h}}$, i.e.,

$$\mathbf{Q}^* = \mathbf{V} \text{diag}((d_1^*)^2, \dots, (d_m^*)^2, 0, \dots, 0) \mathbf{V}^H.$$

Then comes to the scalar power-allocation problem

$$\max_{\{d_i^2 \geq 0\}} \sum_{i=1}^m \log_2 \left(1 + \frac{\lambda_i d_i^2}{\sigma^2}\right), \quad \sum_{i=1}^m d_i^2 \leq P.$$

Since the objective is increasing in every active mode with $\lambda_i > 0$, the power constraint is active. The KKT conditions give, for every active mode,

$$\frac{\lambda_i}{\sigma^2 + \lambda_i d_i^2} = \frac{1}{\nu},$$

where the constant factor induced by $\log_2(\cdot)$ is absorbed into the water level ν . Therefore, $d_i^2 = \nu - \frac{\sigma^2}{\lambda_i}$. Together with the nonnegativity constraint $d_i^2 \geq 0$, this yields (11).

It remains to verify consistency with the I-MMSE necessary condition. Substituting the aligned form of $\mathbf{A}^*$ into the Gaussian MMSE matrix, $\mathbf{E} = \Sigma_{\mathbf{h}} - \Sigma_{\mathbf{h}} \mathbf{A}^H (\mathbf{A} \Sigma_{\mathbf{h}} \mathbf{A}^H + \sigma^2 \mathbf{I})^{-1} \mathbf{A} \Sigma_{\mathbf{h}}$, then every active mode with $d_i^* > 0$,

$$\frac{\lambda_i \sigma^2}{\sigma^2 + \lambda_i (d_i^*)^2} = \frac{\sigma^2}{\nu}.$$

Hence, $\mathbf{A}^* \mathbf{E}^* = \frac{\sigma^2}{\nu} \mathbf{A}^*$, which is exactly (6) with $\mu = 1/\nu$. $\square$

Remark: For Gaussian inputs, the global MI maximizer is PCA+WF. This global maximizer satisfies the fixed-point condition. However, the fixed-point condition itself is only necessary, not sufficient, for global optimality. While the Gaussian input case provides valuable theoretical insight, the resulting closed-form solution relies critically on the Gaussianity assumption and does not extend to general or discrete input distributions. Nevertheless, the necessary optimality condition derived in Section III.A remains valid irrespective of the input distribution.

C. Fixed-Point Iterative Design

Motivated by the general validity of the necessary optimality condition, we now consider the sensing matrix design problem for arbitrary input distributions. In the absence of closed-form solutions, we develop a numerical optimization approach that directly enforces the necessary condition and enables efficient sensing matrix design for general inputs.

Motivated by (10), a fixed-point iteration for sensing matrix optimization can be constructed as

$$\mathbf{A}^{(k+1)} = \frac{\sqrt{P}}{C_k} \left(\mathbf{A}^{(k)} + \alpha \mathbf{A}^{(k)} \mathbf{E}^{(k)} \right), \quad (12)$$

where $\mathbf{E}^{(k)}$ denotes the Minimum Mean Square Error (Minimum Mean Square Error) matrix associated with $\mathbf{A}^{(k)}$, $C_k = \|\mathbf{A}^{(k)} + \alpha \mathbf{A}^{(k)} \mathbf{E}^{(k)}\|_F$ is the norm-controlling factor, and α is an adjustable gradient ascending rate.

For non-Gaussian channel distributions, $\mathbf{E}^{(k)}$ generally **lacks a closed-form expression and must be evaluated numerically**, e.g., via Monte Carlo integration. Although the iteration in (12) is not guaranteed to be a contraction mapping, it provides a practical approach to obtaining stationary points of the MI objective and has been shown to be effective in precoding problems with arbitrary inputs [4].

For arbitrary channel distributions, $\mathbf{E}^{(k)}$ is estimated via Monte-Carlo sampling. Drawing samples $\{\mathbf{h}_i\}_{i=1}^N \sim p(\mathbf{h})$ and $\mathbf{y}_i = \mathbf{A}^{(k)} \mathbf{h}_i + \mathbf{w}_i$, the posterior mean is approximated by importance sampling

$$\hat{\mathbf{h}}(\mathbf{y}_i) \approx \sum_{j=1}^N \mathbf{h}_j \frac{p(\mathbf{y}_i | \mathbf{h}_j) p(\mathbf{h}_j)}{\sum_{\ell=1}^N p(\mathbf{y}_i | \mathbf{h}_\ell) p(\mathbf{h}_\ell)}. \quad (13)$$

The MMSE matrix is then estimated as

$$\mathbf{E}^{(k)} = \frac{1}{N} \sum_{i=1}^N \mathbf{h}_i \mathbf{h}_i^H - \frac{1}{N} \sum_{i=1}^N \hat{\mathbf{h}}(\mathbf{y}_i) \hat{\mathbf{h}}(\mathbf{y}_i)^H. \quad (14)$$

As $N \rightarrow \infty$, this estimator converges to the true MMSE matrix, yielding a consistent approximation of the I-MMSE fixed-point condition. The details are in Algorithm 1.

IV. NUMERICAL RESULTS

This section evaluates the proposed sensing matrix design using both a synthetic discrete-input benchmark and a realistic sample-based dataset [7]. All results are obtained under the upper bound of the sensing power constraint $\|\mathbf{A}\|_F^2 \leq P$ with P set as 1, and additive noise variance as $\sigma^2 = 0.5$.

Algorithm 1 Sensing Matrix Design for General Inputs: Normalized I-MMSE Gradient Ascent for MI-Driven Sensing Matrix Design

Require: Power constraint P , step size α , noise variance σ^2 , number of Monte-Carlo samples N

Ensure: Optimized sensing matrix $\mathbf{A}^*$

- 1: Initialize $\mathbf{A}^{(0)}$ such that $\|\mathbf{A}^{(0)}\|_F^2 = P$
- 2: **for** $k = 0, 1, 2, \dots$ **do**
- 3: Draw samples $\{\mathbf{h}_i\}_{i=1}^N \sim p(\mathbf{h})$
- 4: Generate measurements $\mathbf{y}_i = \mathbf{A}^{(k)}\mathbf{h}_i + \mathbf{w}_i$, $\mathbf{w}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$
- 5: **for** $i = 1, \dots, N$ **do**
- 6: Compute posterior weights $w_{ij} = \frac{p(\mathbf{y}_i|\mathbf{h}_j)p(\mathbf{h}_j)}{\sum_{\ell=1}^N p(\mathbf{y}_i|\mathbf{h}_\ell)p(\mathbf{h}_\ell)}$
- 7: Estimate posterior mean $\hat{\mathbf{h}}(\mathbf{y}_i) = \sum_{j=1}^N w_{ij}\mathbf{h}_j$
- 8: **end for**
- 9: Estimate MMSE matrix

$$\mathbf{E}^{(k)} = \frac{1}{N} \sum_{i=1}^N \mathbf{h}_i \mathbf{h}_i^H - \frac{1}{N} \sum_{i=1}^N \hat{\mathbf{h}}(\mathbf{y}_i) \hat{\mathbf{h}}(\mathbf{y}_i)^H$$

- 10: Update sensing matrix $\tilde{\mathbf{A}}^{(k+1)} = \mathbf{A}^{(k)} + \alpha \mathbf{A}^{(k)} \mathbf{E}^{(k)}$
 - 11: Normalize $\mathbf{A}^{(k+1)} = \frac{\sqrt{P}}{\|\tilde{\mathbf{A}}^{(k+1)}\|_F} \tilde{\mathbf{A}}^{(k+1)}$
 - 12: **end for**
 - 13: **return** $\mathbf{A}^* = \mathbf{A}^{(k)}$
-

A. Experimental Setup and Baselines

We consider the MI as the performance metric. Maximizing MI provides a task-agnostic criterion that characterizes the intrinsic informativeness of the acquired measurements, independent of any specific downstream channel estimation or decoding algorithm. The following baselines are considered:

- **Proposed normalized I-MMSE design:** the proposed necessary-condition-based iteration using empirical MMSE estimation, as in Algorithm 1.
- **Gaussian bound for $\mathbf{A}$:** $\log_2 \det(\mathbf{I} + \frac{1}{\sigma^2} \mathbf{A} \mathbf{K}_h \mathbf{A}^H)$, where $\mathbf{K}_h$ is the empirical covariance of $\mathbf{h}$.
- **Optimal Gaussian covariance bound:** the maximum of the above expression over all feasible $\mathbf{A}$, that is, power-constrained PCA+WF.
- **MWF over covariance eigenmodes [6]:** a classical benchmark allocating sensing power across dominant covariance eigenmodes using empirical MMSE-based KKT conditions.

B. DeepMIMO Dataset Results

We evaluate a realistic sample-based scenario using the DeepMIMO dataset (O1 scenario, where the channel vectors are generated grid-by-grid), where $\mathbf{h} \in \mathbb{C}^8$ represents channel vectors extracted from ray-tracing-based simulations and $m = 4$. The samples are normalized to ensure unit average input power, and all expectations are computed empirically from the dataset.

Fig. 2 presents the results for the DeepMIMO sample-based channel vectors. The fixed-point design consistently improves the empirical MI and converges stably, outperforming the MWF baseline. A large gap to the optimal covariance upper bound is observed, reflecting the strong non-Gaussianity and dependence structure of the DeepMIMO channel distribution.

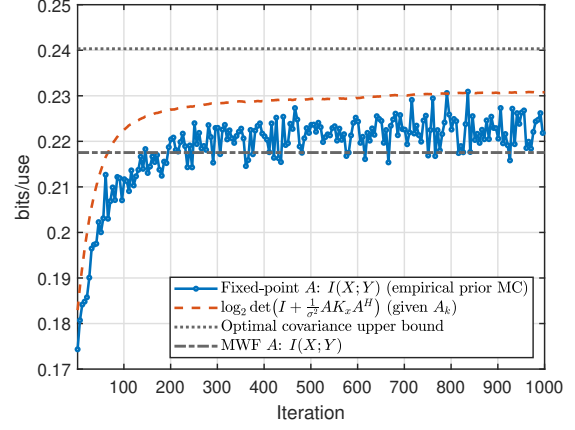


Fig. 2. Sensing matrix iteration for the DeepMIMO dataset ($n = 8, m = 4$).

This gap indicates the intrinsic looseness of covariance-based bounds in realistic sample-driven scenarios.

V. CONCLUSION

This paper studied mutual information-driven sensing matrix design for MISO channel estimation under general input distributions. By deriving a necessary optimality condition based on the I-MMSE relationship, we showed that classical power allocation strategies can be suboptimal for finite or non-Gaussian inputs and that non-diagonal sensing matrices enable improved information extraction. Numerical results with both synthetic and DeepMIMO datasets validated the effectiveness of the proposed design and further revealed the limitations of covariance-based optimality criteria in information-centric channel estimation.

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